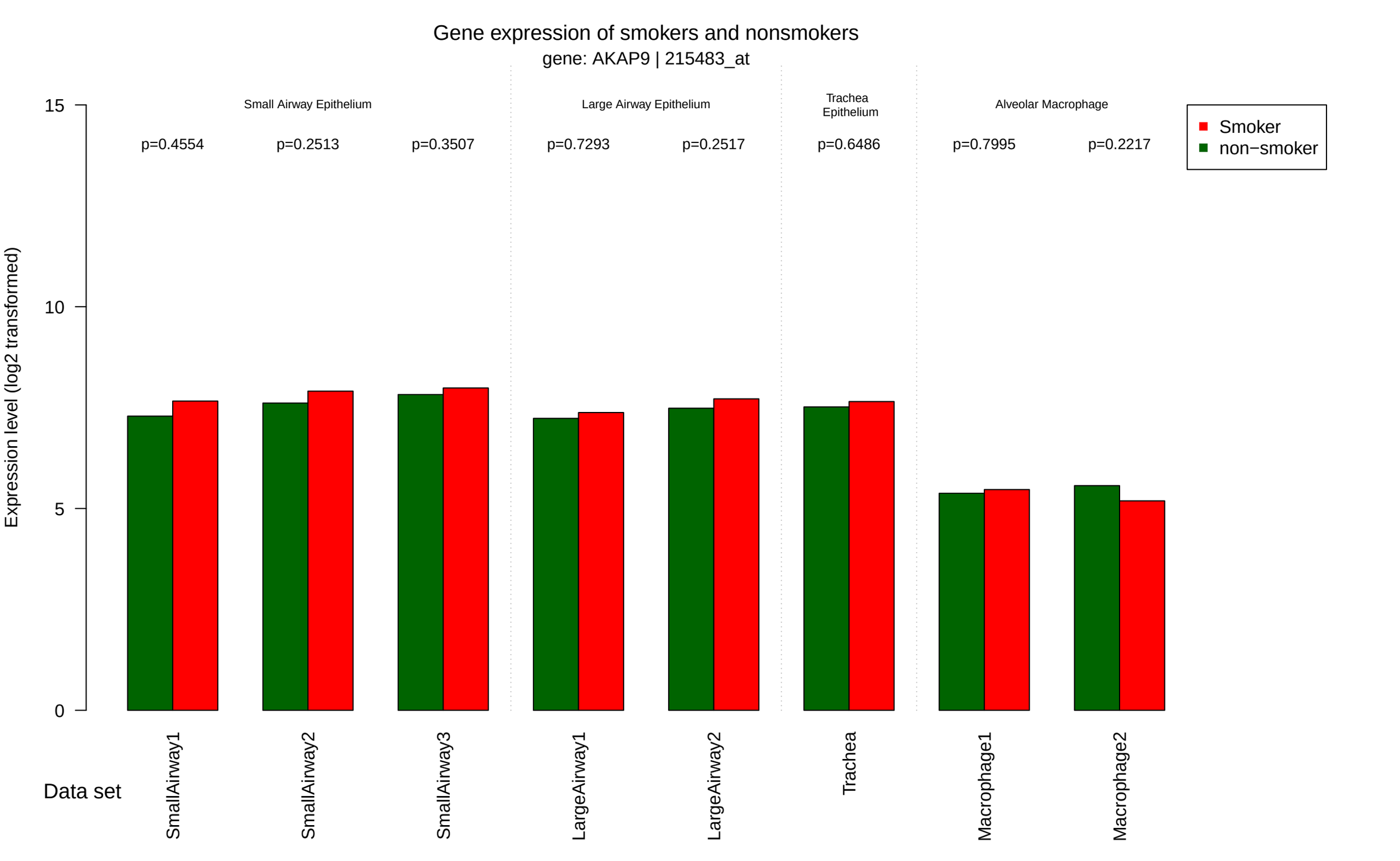
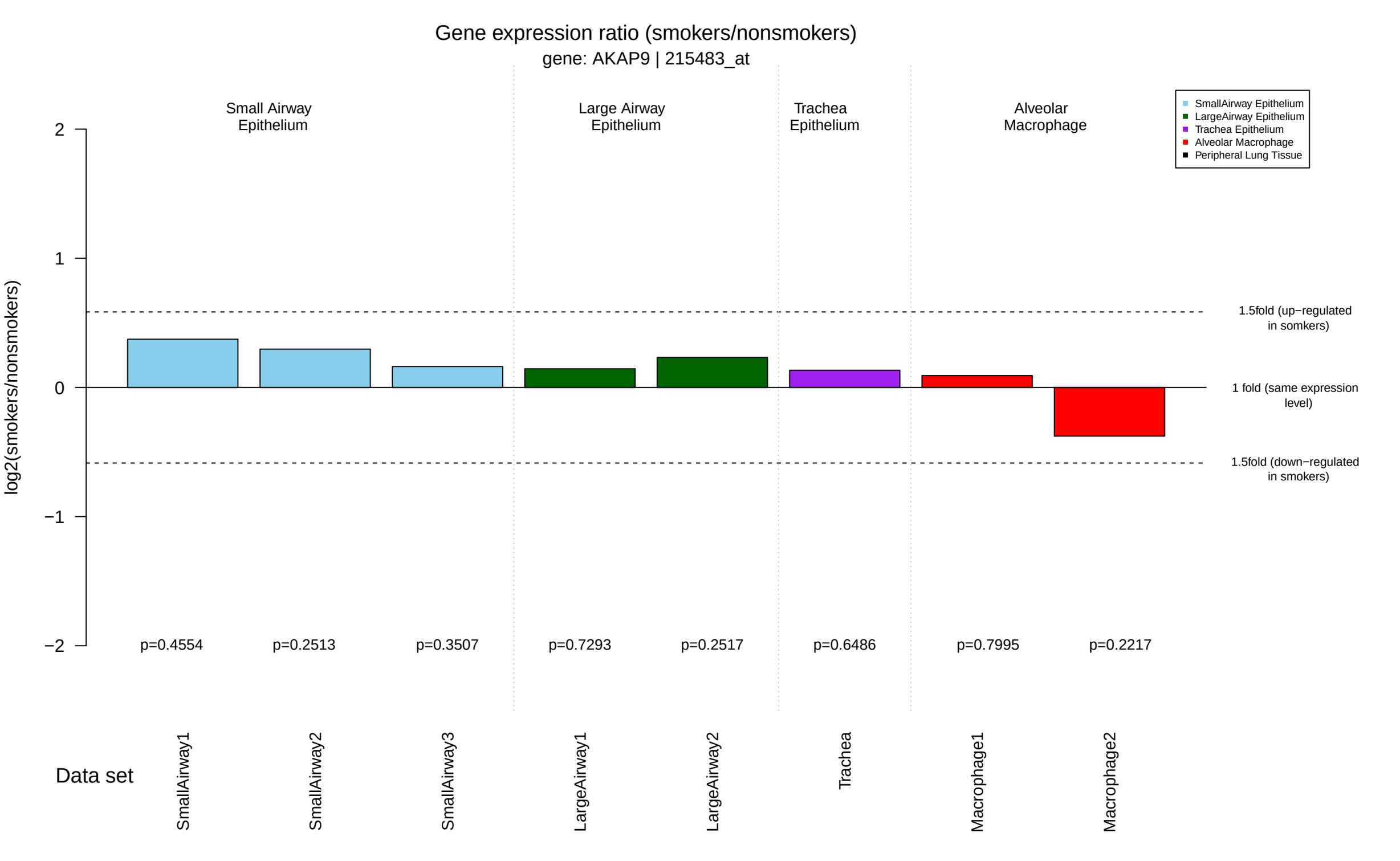
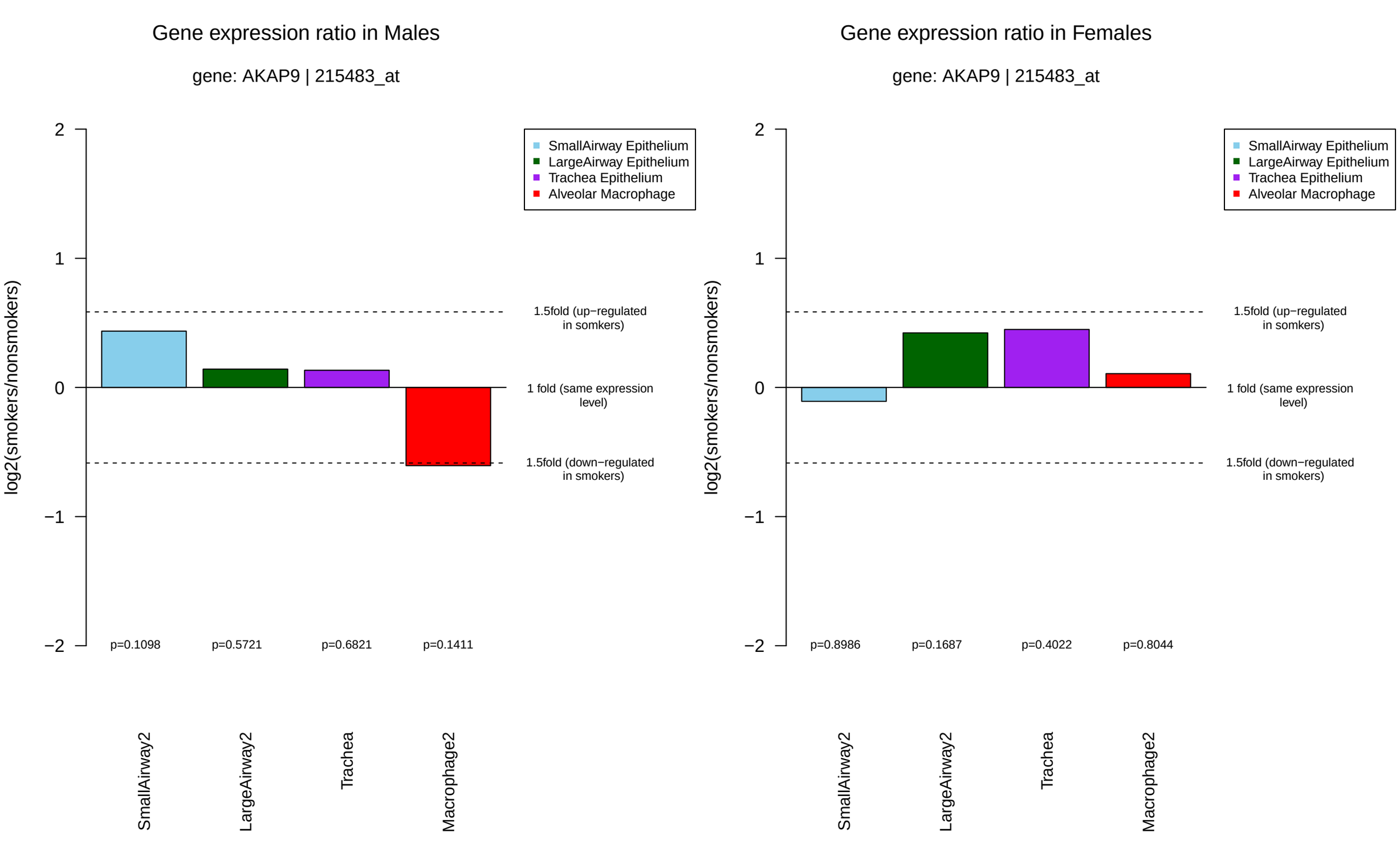
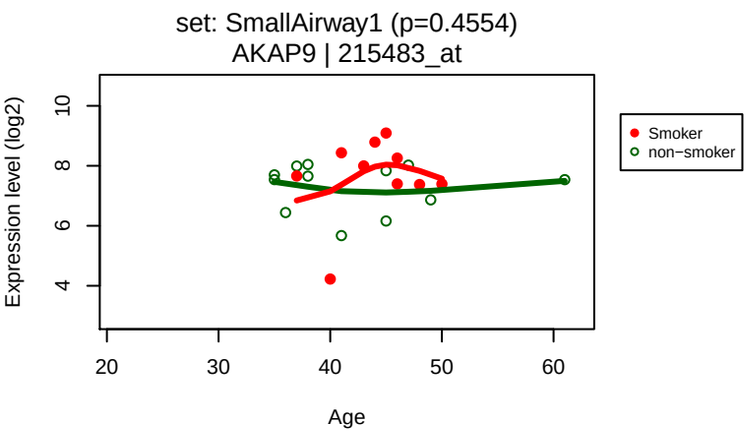
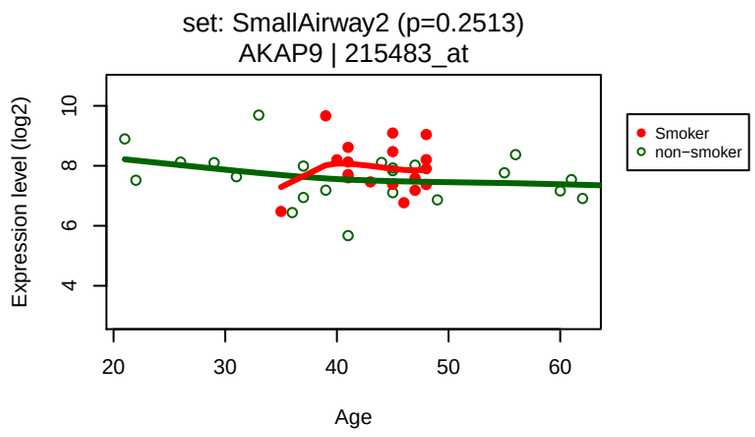
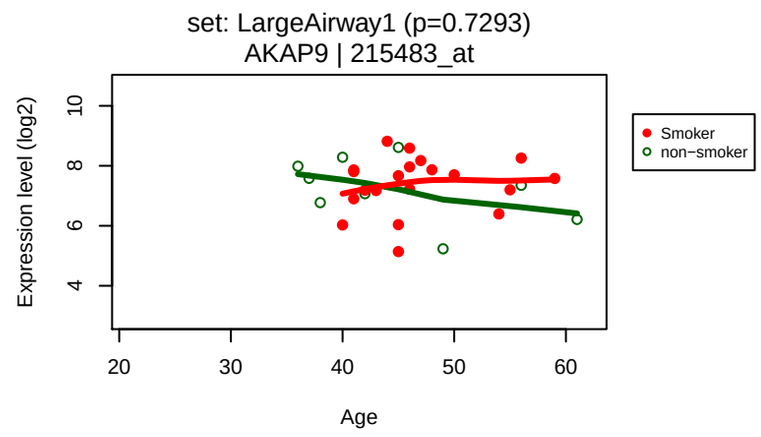

Gene expression of smokers and nonsmokers
gene: AKAP9 | 215483_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4554
p=0.2513
p=0.3507
p=0.7293
p=0.2517
p=0.6486
p=0.7995
p=0.2217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

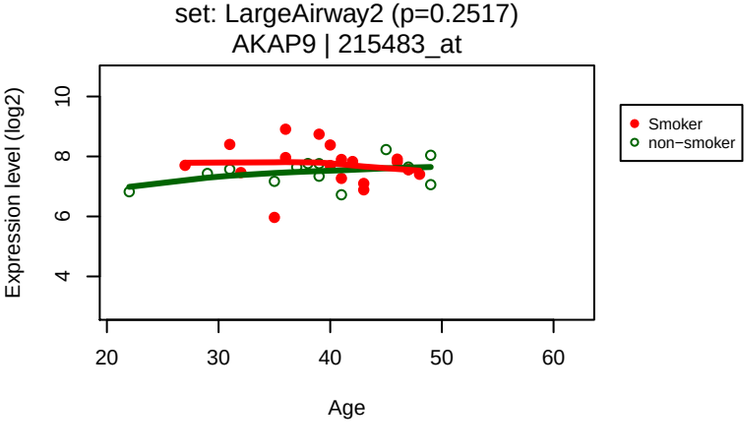

Gene expression ratio (smokers/nonsmokers)
gene: AKAP9 | 215483_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4554
p=0.2513
p=0.3507
p=0.7293
p=0.2517
p=0.6486
p=0.7995
p=0.2217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

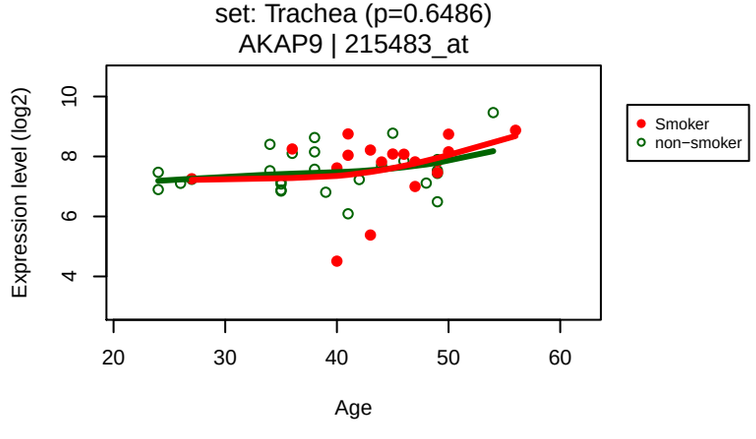

Gene expression ratio in Males
gene: AKAP9 | 215483_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1098
p=0.5721
p=0.6821
p=0.1411
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP9 | 215483_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8986
p=0.1687
p=0.4022
p=0.8044
SmallAirway2
LargeAirway2
Trachea
Macrophage2

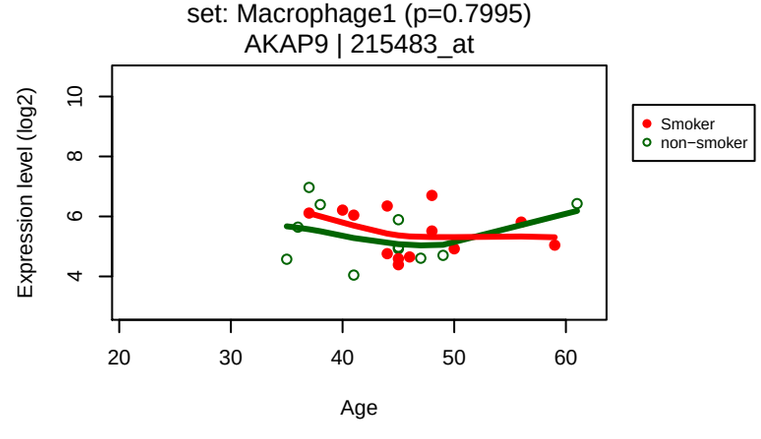

set: SmallAirway1 (p=0.4554)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker

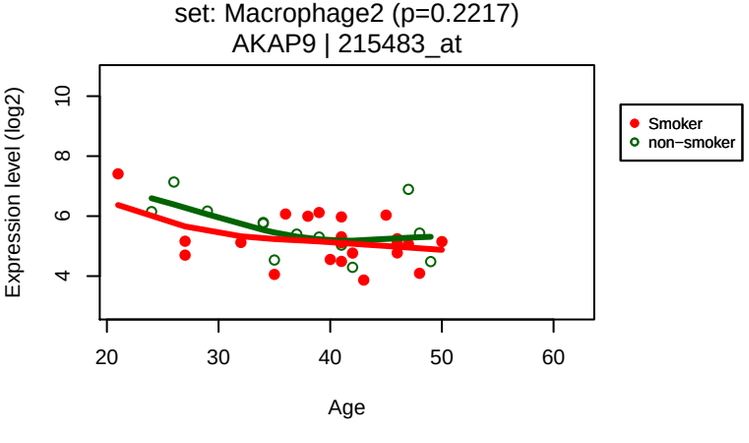

set: SmallAirway2 (p=0.2513)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker

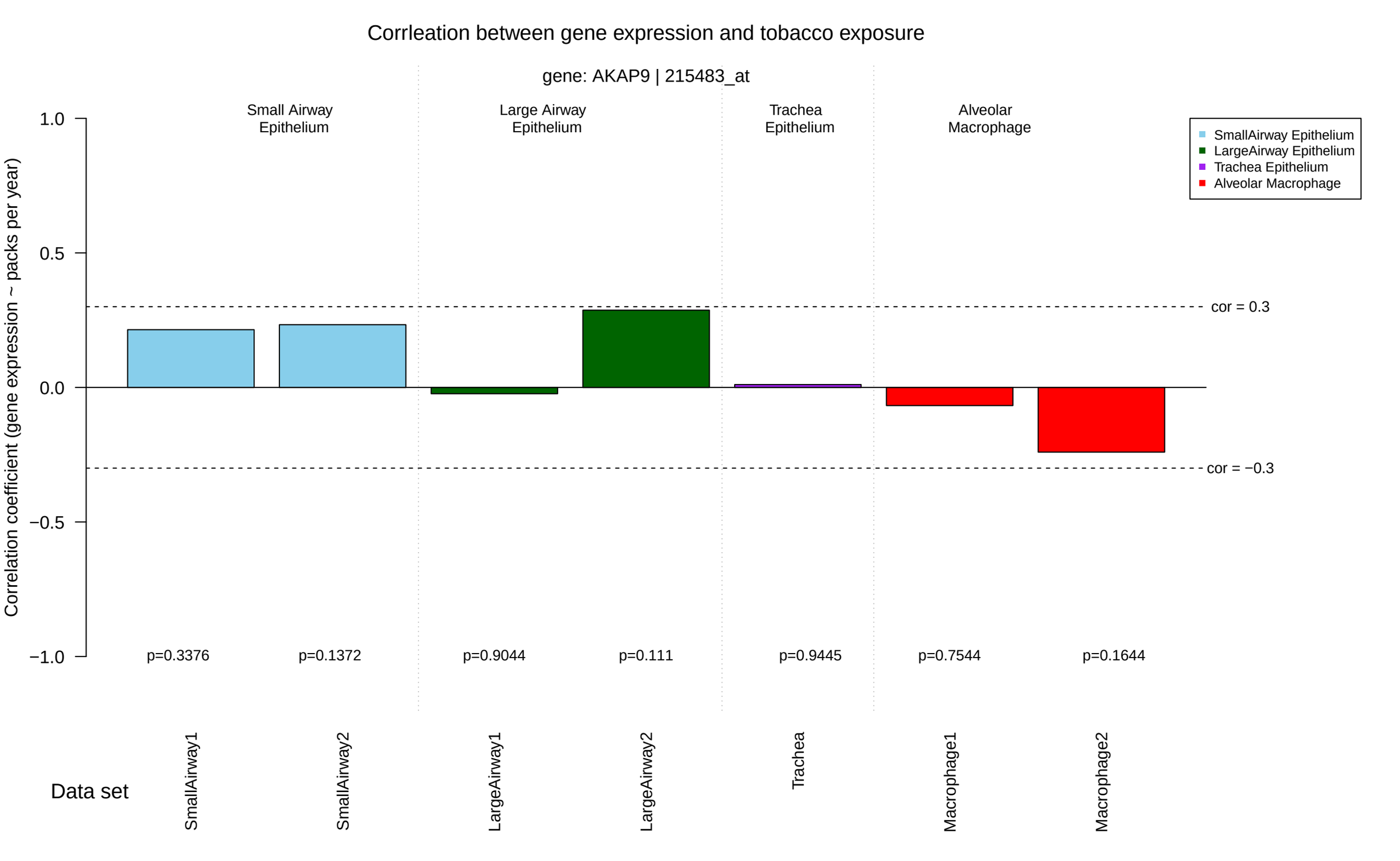

set: LargeAirway1 (p=0.7293)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2517)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6486)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7995)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2217)
AKAP9 | 215483_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP9 | 215483_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3376
p=0.1372
p=0.9044
p=0.111
p=0.9445
p=0.7544
p=0.1644
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2